



FSBI: Deepfake detection with frequency enhanced self-blended images

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ABSTRACT

Advances in deepfake research have led to the creation of almost perfect image manipulations that are undetectable to the human eye and some deepfake detection tools. Recently, several techniques have been proposed to differentiate deepfakes from real images and videos. This study introduces a frequency enhanced self-blended images (FSBI) approach for deepfake detection. This proposed approach utilizes discrete wavelet transforms (DWT) to extract discriminative features from self-blended images (SBI). The features are then used to train a convolutional network architecture model. SBIs blend the image with itself by introducing several forgery artifacts in a copy of the image before blending it. This prevents the classifier from overfitting specific artifacts by learning more generic representations. These blended images are then fed into the frequency feature extractor to detect artifacts that could not be detected easily in the time domain. The proposed approach was evaluated on FF++ and Celeb-DF datasets, and the obtained results outperformed state-of-the-art techniques using the cross-dataset evaluation protocol, achieving an AUC of 95.49% on Celeb-DF dataset. It also achieved competitive performance in the within-dataset evaluation setup. These results highlight the robustness and effectiveness of our method in addressing the challenging generalization problem inherent in deepfake detection. The code is available at <https://github.com/gufranSabri/FSBI>.

1. Introduction

The rise of deepfake technology has introduced a new era of digital manipulation, posing severe threats to the authenticity and integrity of multimedia content. Deepfakes, which are artificially generated media convincingly mimicking real individuals or events, present several critical challenges. One major concern is their potential to spread misinformation on an unprecedented scale, undermining public trust in digital content and traditional media. They can be weaponized to create fake news, manipulate public opinion, and interfere with political processes, such as elections or policy decisions. Deepfakes also enable malicious content creation, such as non-consensual explicit videos or fraudulent communications, which can lead to reputational damage, emotional distress, and legal complications for victims. Moreover, deepfake technology threatens privacy and security by facilitating identity theft and impersonation in financial fraud or cyberattacks. These multifaceted challenges underscore the urgent need for robust detection methods to mitigate the risks posed by this rapidly evolving technology.

Several techniques have been proposed in the literature for generating fake images that can be synthesized entirely or partially. Artificial neural networks [1], auto-encoding variations [2,3], and generative adversarial networks (GAN) [4] play significant roles in the development

and creation of modified and fake images. More realistic fake images have been created using GAN variants [5], such as Progressive GAN [6], StyleGAN [7], and SNGAN [8]. Techniques such as Face Swap [9], Neural Textures [10], and Face2Face [11] have been proposed for real-time face manipulations.

The detection of deceptive digital creation is a complex and pressing challenge. This complexity arises from the rapid advancements in generative models, which produce content that is increasingly indistinguishable from authentic media. Furthermore, the diversity of manipulation techniques and the availability of tools for creating deepfakes make it difficult to develop universally effective detection strategies. Traditional methods for identifying deepfakes have become increasingly ineffective as the sophistication of generative models and manipulation techniques continues to evolve. In this context, innovative and robust deepfake detection mechanisms are imperative for safeguarding the credibility of digital media and maintaining public trust.

In the dynamic field of deepfake detection, diverse techniques that combine traditional methods with cutting-edge deep learning approaches have emerged. Traditional techniques depend on the examination of facial expressions, eye movements, and lip synchronizations to detect inconsistencies. In addition, visual facial artifacts such as eyes lacking realistic reflections and vibrance or teeth appearing as white

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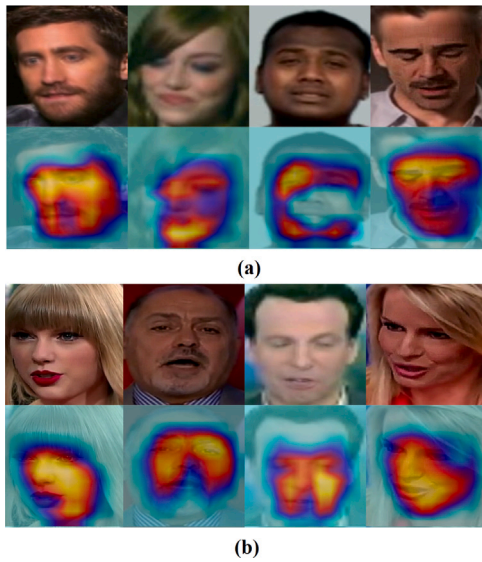


Fig. 1. Examples of the artifact regions detected by the proposed FSBI approach on samples from (a) Celeb-DF and (b) FF++ datasets.

blobs, have been exploited for detection [12–15]. These artifacts can be easily captured in high-resolution images; however, downsampling the image size to meet computational constraints usually creates difficulties in capturing these artifacts and sometimes introduces resolution inconsistencies. Therefore, some researchers have used preprocessing techniques to address this issue and enhance the generalization of the model by capturing a range of visual artifacts [16,17]. However, traditional deepfake detection techniques are less effective for deepfake generation algorithms. Recently, researchers have leveraged deep learning techniques to scrutinize patterns in facial features, lighting, and background elements to detect deepfakes. Zhao et al. proposed a multi-attentional deepfake detection network that incorporated regional independence loss and attention-guided data augmentation [18]. Sun et al. [19] incorporated LRNet as a novel deepfake detection approach, whereas Zhao et al. [20] improved accuracy by integrating anti-deepfake labels into facial identity features. Recent works incorporated deep learning with feature enhancement for deepfake detection. Wang et al. [21] used a Wavelet-guided AutoEncoder and Texture-Content Attention Maps Learning to enhance forgery detection with high-frequency details. Zheng et al. [22] introduced DT-TransUNet, a dual-task network that performs both detection and segmentation. The proposed approach utilizes a Multi-Scale Spatial Frequency Feature (MSSFF) module with Stationary Wavelet Transform (SWT) to improve texture feature extraction.

Shiohara et al. [23] proposed deepfake data synthesized using a self-blended images (SBIs) approach. The pseudo-source and target images were generated for each original image by applying several transformations. These images were blended to create SBIs used for deepfake detection. This study proposes an enhancement to the approach proposed in et al. [23]. We propose a frequency-enhanced SBIs (FSBIs) approach for deepfake detection. This approach extends the SBI method by extracting features from the SBIs in the frequency domain to develop a more generic and robust detector. Each channel of the blended images resulting from the SBI was fed into discrete wavelet transforms (DWT). The resulting features were combined and fed into a Convolutional Neural Network (CNN) model to learn the artifacts in deepfake images. We used within-dataset and cross-dataset evaluation protocols to evaluate the proposed approach and new state-of-the-art results were obtained using the FF++ and Celeb-DF datasets. The proposed approach showed a good capability for detecting the artifacts on the deepfake images, as shown in Fig. 1. This study makes the following contributions:

- Proposing an innovative approach for deepfake detection.
- Discussing the efficiency of frequency-based transforms for detecting artifacts on deepfake images.
- Evaluating the generalization of the proposed approach by performing cross-dataset evaluation.
- Reporting a new state-of-the-art performance in the Celeb-DF dataset.

The remainder of this paper is organized as follows: Section 2 surveys the literature on deepfake detection and Section 3 presents the framework of the proposed approach. Section 4 presents the study's experiments and results along with comparisons and discussions. Finally, Section 5 concludes the study.

2. Related works

Several techniques have been proposed for deepfake detection. These techniques vary from simple computer vision algorithms to complex deep-learning neural networks. Therefore, we classified these techniques into traditional and deep-learning approaches.

Traditional Techniques. These techniques depend mainly on classic computer vision operations for artifact detection. Although the term “traditional techniques” may suggest that these techniques are outdated, they are still being used for deepfake detection. Jeong et al. [24] employed bilateral high-pass filters to detect amplification effects on fake images. Similarly, Liu et al. proposed by a deepfake detector that utilizes temporal identity inconsistencies between human faces in video frames [25]. Although these techniques accurately detect deepfakes, significant advances in deepfake generation have motivated researchers to search for alternative deep learning techniques [26].

Deep Learning Techniques. Most recent deepfake detection techniques depend on neural networks [27]. The majority of these techniques [28–33] utilize CNNs for learning and classifying images into real and fake classes. Most of these CNN models do not undergo end-to-end training; instead, they rely on pre-trained models for feature extraction. EfficientNet models are the dominant pre-trained models used in most CNN-based deepfake detection techniques [34–36]. To boost the accuracy of these techniques, several researchers have utilized CNNs with transformer models [23,34–38]. Wodajo et al. [38] used a CNN to extract spatial features and fed them into a vision transformer (ViT) model that used an attention mechanism to categorize them into real or fake images. Similarly, CNNs have been combined with ViT by other researchers [35–37,39] and good results were obtained. Chen et al. [40] proposed a face forgery detection method that learns local features and detects inconsistencies. To address the generalizability of current deepfake detection methods, Chen et al. [41] employed a self-supervised learning-based method. The model employs a pool of forgery configurations and synthesized augmented forgeries to detect forgeries in images and videos. Lee et al. [42] presented a digital forensic tool for detecting different types of deepfakes. The proposed tool learns features using an AE-based model.

The generalization of deepfake detection techniques is essential for the development of real-time systems. Therefore, researchers typically perform cross-dataset evaluations with their proposed deepfake detection approaches [5,29–32,35,43]. In this setting, the technique is evaluated on a dataset different from that used for model training. For example, Coccomini et al. [35] performed cross-forgery analysis using ViT to detect deepfakes in various datasets. Luo et al. [30] used high-frequency features and CNNs to generalize the detection methods. The MCX-API [31] method employs pairwise learning to improve model generalization. Deepfake Disrupter [43] was used with the help of human input and deepfake detection tools to identify deepfakes. However, most proposed techniques cannot detect deepfakes that are not present in the training model [26].

Recent deep learning advancements in deepfake detection have focused on tackling spatio-temporal inconsistencies and dynamic motion

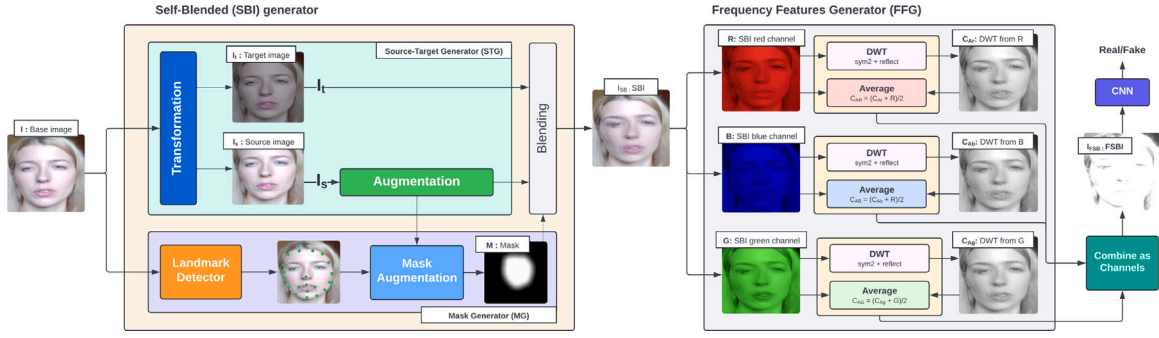


Fig. 2. Framework of the FSBI model. This SBI is then decomposed into R, G, and B channels by the FFG. DWTs are computed for each channel individually, and the approximate coefficients are obtained. These approximate coefficients are then combined with the original channel through a simple averaging operation. The channels are stacked channel-wise to form the final FSBI image. The resulting FSBI images are used to train a CNN classifier to recognize real and fake images.

analysis. Dynamic Difference Learning (DDL) utilizes long and short-range inter-frame motion analysis with modules like DFDC and MSA to distinguish facial motion from manipulation artifacts and model spatiotemporal inconsistencies [44]. Complementary Dynamic Interaction Networks (CDIN) analyze global and local dynamics, particularly in the mouth region, using a Complementary Cross Dynamics Fusion Module (CCDFM) to enhance generalization to unseen forgery techniques [45]. Head Pose Estimation (HPE) patterns have demonstrated robust detection across various datasets by analyzing temporal pose patterns [46]. Further advancements in feature enhancement include Wang et al. [21], who introduced a Wavelet-guided AutoEncoder and Texture-Content Attention Maps to improve high-frequency detail detection, and Zheng et al. [22], who proposed DT-TransUNet, a dual-task network employing a Multi-Scale Spatial Frequency Feature (MSSFF) module and Stationary Wavelet Transform (SWT) for improved texture feature extraction.

Addressing the challenges of generalization and fairness, recent works have emphasized overcoming biases in detection accuracy across demographic groups. Fairness-aware loss functions and disentanglement learning approaches have been introduced to ensure demographic and domain-agnostic detection, enabling more equitable performance across diverse datasets [47,48]. Additionally, Xu et al. [49] tackled generalization issues by proposing a reconstruction-classification network that learns facial content and gradients separately, fuses these features, and categorizes forged faces using a Bi-Directional attention module for reconstruction disparity.

3. Frequency enhanced self-blended images (FSBI)

We propose an FSBI approach for deepfake detection, and its framework is given in Fig. 2. The proposed approach comprises three modules: an SBI generator, a frequency feature generator (FFG), and a CNN classifier. The original image is fed into the SBI generator component to create a blended image with artifacts. The FFG applies a series of DWT wavelets to the resulting blended image to extract features in the frequency domain. The features are then fused and fed into a CNN classifier trained with an EfficientNet-B5 pre-trained model to learn and classify images as real or fake.

3.1. Self-blended image (SBI)

SBI images are generated when an image is blended with itself. It was proposed by Shiohara et al. and creates deepfakes by blending a face with its mask, which is generated by applying several transformations SBI [23]. The SBI method uses three primary stages to generate deepfakes from an original image. First, a source-target generator (STG) component takes an image as an input and generates pseudo-source (I_s) and target (I_t) images. These images are generated by performing random image transformations to the original image. These transformations involve manipulating the values of RGB and HSV channels and

changing the brightness and contrast of the image. Additional augmentation operations involve downsampling and translation is applied to the source image to create landmark mismatches.

In the second stage, the SBI uses a mask generator (MG) to create a gray-scale mask image that blends both the source and target images. The MG generates a mask by detecting the landmarks of the facial region and calculating the convex hull from these detected landmarks. For this process, the SBI method adopts the FaceXray BI [50] to perform landmark detection followed by elastic deformation [31]. The masks are smoothed twice using two Gaussian filters and eroded after the first filter is applied.

Finally, the SBI blends the source and target images using the generated mask. The blended image is created by eroding the source image with the mask generated by the MG module. The resulting eroded image is then combined with the eroded target image, using the inverse of the mask, to produce the final SBI. The steps were as follows:

$$I_{SBI} = I_s \odot M + I_t \odot (1 - M), \quad (1)$$

where I_s is the source image, I_t is the target image, and M is the generated mask.

3.2. Frequency features generator (FFG)

The blended image obtained from the SBI module was fed into the FFG module. This module leverages DWT to extract more discriminative features from blended images, aiding in the detection of artifacts in deepfake images. DWT is a signal- and image-processing technique that decomposes a signal into different frequency components by iteratively applying filters to the signal, dividing it into approximate and detailed coefficients at various resolutions. This process provides multiresolution analysis that captures both global and local features within the image.

DWT families, such as Haar, Daubechies, Symlet, Biorthogonal, and Coiflet, offer different properties suitable for various applications. Haar is computationally efficient but does not perform well for complex structures. Daubechies, with varying orders, are effective for fine details and compression. Symlet balances smoothness and oscillations, making it suitable for signals with both smooth and abrupt changes. Biorthogonal offers flexibility with separate sets for decomposition and reconstruction, while Coiflet, which has improved regularity, is suitable for applications requiring smoother wavelets. The choice between these families depends on the specific characteristics of the deepfake images and the requirements of the detection task. In this study, we evaluated the Haar, Symlet, and Biorthogonal wavelet types for deepfake detection and obtained the best results using the Symlet wavelet type.

Additionally, several signal extension modes can be applied to reduce artifacts resulting from signal extrapolation when computing the DWT. These extension modes include symmetric, reflective, and

antireflective modes. We explored all these modes in this study and achieved the highest score using the reflective mode.

The RGB channels of the SBI were decomposed into three channels (R, G, and B), and the DWT coefficient was computed for each channel. The computed approximate coefficient for each channel was resized to match the original images and averaged with the corresponding SBI channel. The resulting coefficients were concatenated depthwise to create a three-channel DWT image, referred to as FSBI. This image highlights all the possible artifacts introduced into the deepfake image during its generation.

3.3. CNN model

The final component of the proposed FSBI approach is a CNN model trained to detect artifacts in the FSBI and classify images as real or fake. This model was trained on FSBI, which encode the SBI using frequency features extracted from each channel. We fine-tuned the EfficientNet-B5 [51] pre-trained on ImageNet, for this task. EfficientNet-B5, a member of the EfficientNet family, features compound scaling, where width, depth, and resolution are uniformly scaled to balance model capacity and computational cost. This design prioritizes state-of-the-art accuracy in image classification tasks, such as those on ImageNet, while using fewer parameters compared to other models with similar performance.

4. Experiments

4.1. Experimental setup

Datasets. We employed two prominent datasets to evaluate the proposed approach: FF++ [52] and Celeb-DF [53]. The FF++ dataset consists of 1,000 videos with approximately 500,000 frames. This dataset has fake images created using deepfake (DF) [54], face swap (FS) [55], neural textures (NT) [56], and Face2Face (F2F) [57] methods. From the 1,000 real videos, FF++ constructed over 1.8 million images derived from 4,000 fake videos. Due to the large size of the FF++ dataset, we used the compressed c23 version of the FF++ dataset (FF++(LQ)) in this work. We also used the Celeb-DF (version 2) dataset, which comprises large-scale deepfake videos generated using specialized synthesis methods. The dataset comprises 5,639 high-quality deepfake videos featuring celebrities, which amount to approximately two million frames.

Implementation Details. The model was trained and evaluated using an Ubuntu 22.04 LTS machine equipped with an NVIDIA RTX A5000 GPU with 48 GB of memory. The PyTorch framework was used to implement the proposed approach. The proposed model was trained using an early stopping technique to avoid overfitting. However, to obtain a valid comparison with the baseline model, the SBI method [23], we used the same number of epochs (100) to train the SBI method.

Preprocessing. The preprocessing pipeline applies a range of augmentations to enhance model robustness. For source images, transformations like RGB shift, hue, saturation, brightness, and contrast adjustments are applied, along with downscaling and sharpening. Additional augmentations, including image compression, are applied to both source and target images. These transformations simulate real-world variations and improve the model's generalization.

Training. For the CNN model, we fine-tuned the EfficientNet-B5 model that was pre-trained on ImageNet. We employed the Sharpness-Aware Minimization (SAM) optimizer to optimize the training process. We set the momentum for the optimizer to 0.9 and the learning rate to 0.001. The learning rate linearly decays after 75 epochs. We train the model on a batch size of 16. The size of the image is set to 380×380 .

Inference. During the inference stage, a set consisting of 32 frames was extracted from each video. If the frame contains two or more faces,

Table 1

The AUC of the FSBI approach with within- and cross-dataset evaluations.

Evaluation	Train	Test	Manipulation				Average
			DF	FS	F2F	NT	
Within-dataset	FF++	FF++	99.45	97.09	95.37	88.59	95.13
	Celeb-DF	Celeb-DF					95.40
Cross-dataset	Celeb-DF	FF++	96.76	95.43	87.62	78.07	89.47
	FF++	Celeb-DF					95.49

the classifier assessed each face, and the confidence associated with the highest fakeness was considered the predicted confidence for that frame. After obtaining the predictions for all the frames, we calculated the average to determine the video's overall predictions. To ensure a fair comparison, we evaluated all videos in the test sets, setting the confidence to 0.5 for videos when no faces were detected in any frame.

4.2. Results

In this section, we present the evaluation results of the proposed FSBI approach for deepfake detection. Two settings were used to evaluate the proposed approach: within-dataset and cross-dataset. The Area Under the Curve (AUC) metric was used in both settings.

Within-dataset Evaluation. In this evaluation, the performance of the FSBI was evaluated using the same dataset on which it was trained. Table 1 presents the results obtained using the FF++ and Celeb-DF datasets. Deepfake images of the FF++ dataset are available in four formats based on the deepfake generation technique (DF, FS, F2F, and NT), as discussed in Section 4.1. We evaluated the performance of FSBI on all versions of FF++. Our model reported high AUC for both datasets, as indicated at the top of Table 1. For FF++, the FSBI approach can recognize deepfakes generated using almost all manipulation methods. A high AUC was obtained for the FF++ dataset when deepfake images were generated using the DF method. In contrast, a low AUC was obtained when the NT manipulation technique was used. This could be attributed to the fact that the NT method introduces a novel paradigm for image synthesis. NTs are learned feature maps that store high-dimensional information on 3D mesh proxies, which differ significantly from the traditional textures used in other deepfake generation techniques. This unique representation presents challenges for deepfake detection models, reducing their effectiveness. In addition, the compressed videos of the FF++ dataset (FF++(LQ)) had low resolution. Detecting deepfakes in low-resolution images is more challenging than that in raw high-resolution images. For the Celeb-DF dataset, a high AUC value was obtained when the approach was trained and evaluated using samples from the same dataset.

Cross-dataset Evaluation. A crucial aspect of assessing the generalizability of our approach is to evaluate its performance across different data distributions and manipulation techniques. This type of evaluation is called a cross-dataset evaluation and serves as a robust test to gauge the effectiveness and adaptability of any deepfake detection approach. In this evaluation, the dataset used to test the model differed from that used for training. We evaluated the FSBI using the evaluation settings, and the results are presented in Table 1. Notably, when the model was trained on the FF++ dataset and tested on the Celeb-DF dataset, it outperformed related methods, achieving an impressive AUC of 95.49%. Furthermore, training the model on the Celeb-DF dataset yielded satisfactory results when evaluated using the FF++ dataset. The model was evaluated on the FF++ dataset using various manipulation techniques. The table shows that FSBI achieved a high AUC for the deepfakes generated using the DF and FS methods. However, the AUC scores for the FSBI model using the F2F and NT methods were relatively low. The results with NT manipulations aligned with those obtained in the within-dataset evaluation.

Table 2

Our FSBI method demonstrates higher performance compared to other deepfake detection approaches. This is evident when we train and test on the same dataset (Within-DS: Within-dataset) and when trained on one dataset and evaluated on a different dataset (Cross-DS: Cross-dataset).

Eval.	Method	Train	Test Dataset	
			FF++	Celeb-DF
Within-DS	Jevnisek et al. [58]	FF++	75.57	–
	HPE [46]	FF++	86.25	–
	ViT [59]	FF++	95.10	–
	MCX-API [60]	FF++	99.6	–
	CNN+ViT [61]	FF++	91.50	–
	CDIN [45]	FF++	97.80	–
	DFDC+MSA [44]	FF++	96.65	–
	FSBI (ours)	FF++	95.13	–
Cross-DS	MCX-API [60]	Multiple	–	90.87
	FRDM [62]	Multiple	–	79.40
	PCL + I2G [63]	Multiple	–	90.03
	LRL [18]	Multiple	–	78.26
	DFDC+MSA [44]	FF++	–	71.36
	LipForensics [64]	FF++	–	82.40
	FTCN [65]	FF++	–	86.90
	SBI (FF++(HQ)) [23]	FF++	–	93.18
	SBI (FF++(LQ)) [23]	FF++	–	92.87
	RLGC [49]	FF++	–	91.20
	CDIN [45]	FF++	–	85.80
	FSBI (ours)	FF++	–	95.49



Fig. 3. Deepfake samples from the Celeb-DF dataset. The first row contains samples that were deepfakes that were correctly detected by the FSBI approach and not detected by the SBI approach. The second row contains samples that were not detected by both approaches.

4.3. Ablation study

Ablation studies were conducted to verify the effectiveness of the proposed FSBI. We systematically varied the key components and hyperparameters in our deepfake detection pipeline to understand their impact on performance while optimizing the accuracy of the proposed model. The components and hyperparameters were FFG and DWT type and mode, image size, and a pre-trained model, respectively. We conducted ablation studies in the cross-dataset evaluation setting using the FF++ and Celeb-DF datasets for training and testing, respectively.

FFG Component. Our approach consists of three components: the SBI, FFG, and CNN. A specific training procedure was implemented to evaluate the effectiveness of the first two components. The model was trained for 75 epochs on one component and fine-tuned for 25 epochs on the other. We also trained and fine-tuned the model on both components, as shown in Table 3, where half of the images were generated using SBI, and the other half contained images generated using FSBI. The objective is to evaluate the performance of the proposed approach with and without an FFG module. The table shows that training and fine-tuning the model solely on the FSBI resulted in the highest AUC score of 95.49%. The results indicate an overall enhancement in model performance when trained on FSBI compared to SBI alone or when images generated by SBI are used in either training or fine-tuning the

Table 3

Results from experimenting with the effectiveness of the framework components. Model was trained for 75 epochs and finetuned for 25 epochs.

Training		Fine-tuning		AUC (%)
SBI	FSBI	SBI	FSBI	
✓	✗	✓	✗	94.66
✓	✗	✗	✓	94.78
✗	✓	✓	✗	94.01
✗	✓	✗	✓	95.49
✓	✓	✓	✗	94.21
✓	✓	✗	✓	94.07
✓	✓	✓	✓	94.90

Table 4

The performance of the FSBI with different image sizes.

Image size (px)	AUC
170 × 170	88.55
230 × 230	92.55
380 × 380	95.49
512 × 512	91.59

model. This improvement can be attributed to the nature of the FFG, which employs DWT to amplify the frequency of information in the SBI images. Training the model on images generated using the FSBI focuses on the frequency artifacts present in deepfakes. Additionally, DWT introduces texture analysis-based features that are crucial for detecting deepfake artifacts resulting from sharp intensity changes during the generation process.

DWT Variables. DWT plays a crucial role in the FSBI approach for representing and extracting features from SBI images in the frequency domain. Three main wavelet types are available for DWT: Biorthogonal, Symlet, and Haar, and the choice among them depends on the downstream task. Additionally, three signal extension modes—Symmetric, Reflective, and Antireflective—can be applied to reduce artifacts resulting from signal extrapolation during DWT computation. To evaluate the impact of wavelet type and signal extension modes on FSBI performance, we conducted a cross-dataset evaluation, and the results are shown in Table 5. The results reveal that the choice of DWT variables significantly influences the model's performance, with the best configuration being the Symlet wavelet and Reflective mode (M5 model), achieving an AUC of 95.49%. This underscores the importance of carefully selecting DWT parameters to optimize the model's ability to detect deepfake images. Notably, all DWT configurations outperformed the state-of-the-art results on the Celeb-DF dataset, supporting our hypothesis about the effectiveness of DWT for deepfake detection.

Image Size Selection. The choice of image size is crucial for FSBI performance. We experimented with four image sizes: 170×170, 230 × 230, 380×380, and 512 × 512. As shown in Table 4, notable differences in AUC scores were observed with varying image resolutions. Performance improved with larger image sizes, as expected, due to more visible artifacts. However, the model's performance deteriorated with the largest image size of 512 × 512 pixels, likely due to information loss during interpolation from 256 × 256 to 512 × 512.

Choice of the pretrained Model. While the CNN model used the pre-trained EfficientNet-B5 in the FSBI framework, we also evaluated other network architectures. Our experiments included EfficientNet-B4 [51], EfficientNet-B5, ResNet50 [66], and MobileNetV2 [67]. We also present standard SBI results using these models. As shown in Table 6, EfficientNet-based models consistently outperformed ResNet50 and MobileNetV2 across various configurations. EfficientNet-B5 improved the performance of SBI and FSBI by more than 1% compared to EfficientNet-B4, demonstrating that selecting the appropriate network architecture is vital for enhancing deepfake detection accuracy.

Table 5
Ablation study results with different DWT wavelet types and modes.

Model	Wavelet			Mode			AUC
	Biorthogonal	Symlet	Haar	Symmetric	Reflect	Antireflect	
M_1	✓	✗	✗	✓	✗	✗	93.42
M_2	✓	✗	✗	✗	✓	✗	93.97
M_3	✓	✗	✗	✗	✗	✓	93.97
M_4	✗	✓	✗	✓	✗	✗	94.87
M_5	✗	✓	✗	✗	✓	✗	95.49
M_6	✗	✓	✗	✗	✗	✓	95.21
M_7	✗	✗	✓	✓	✗	✗	94.65
M_8	✗	✗	✓	✗	✓	✗	94.65
M_9	✗	✗	✓	✗	✗	✓	94.65

Table 6
The performance of SBI and FSBI with different pre-trained models. Our proposed model consistently outperformed SBI for each pre-trained model.

Models	ResNet-50	MobileNet-V2	EfficientNet-B4	EfficientNet-B5	AUC
SBI [23]	✓	✗	✗	✗	83.04
FSBI (Ours)	✓	✗	✗	✗	84.68
SBI [23]	✗	✓	✗	✗	78.36
FSBI (Ours)	✗	✓	✗	✗	81.72
SBI [23]	✗	✗	✓	✗	93.10
FSBI (Ours)	✗	✗	✓	✗	94.29
SBI [23]	✗	✗	✗	✓	94.66
FSBI (Ours)	✗	✗	✗	✓	95.49

4.4. Comparisons

This section compares our FSBI approach with SOTA deepfake detection techniques. In this comparison, two evaluation settings were used: within-dataset and cross-dataset, as shown in Table 2. For within-dataset evaluation, FF++ was used to train and test the models. Most deepfake detection techniques use this dataset for within-dataset evaluations. To the best of our knowledge, the Celeb-DF dataset has been used in the literature for cross-dataset evaluation.

We also conducted a cross-dataset evaluation using Celeb-DF as the testing dataset. Unlike approaches that use multiple datasets for training [18,60,62,63], our approach uses only the FF++ dataset. To ensure a fair comparison with the SBI approach [23], we reproduced their results using the compressed c23 version (FF++(LQ)) of the FF++ dataset. As shown in Table 2, the proposed approach significantly outperformed all other techniques. In addition, it outperformed the approaches trained on multiple datasets.

To visualize the qualitative differences between the proposed FSBI approach and the SBI method [23], t-SNE was applied to the feature vectors from the last layer of both models. Fig. 4 illustrates the feature space visualizations of the FSBI and SBI methods when trained on the FF++ and Celeb-DF datasets. As shown in the feature visualization plot, the FSBI method generated a distinct boundary between the clusters of the two classes for the FF++ dataset. In contrast, the SBI method exhibited less well-defined boundaries between the two classes. For the Celeb-DF dataset, the FSBI method demonstrated significantly reduced overlap between the two clusters compared to the SBI method. Fig. 3 shows some deepfake samples that were correctly detected by our approach, whereas the SBI method failed to detect them. The figure also includes other samples that were challenging for both methods.

5. Conclusion

In this work, we present a novel deepfake detection method, ESBI, which significantly outperforms existing methodologies. Our approach leverages the synergy of SBI and DWT, enabling our model to detect subtle artifacts in manipulated content. Through extensive experiments and ablation studies, we demonstrate the effectiveness of our proposed method, achieving a remarkable AUC of 95.49%. The strategic combination of FSBI with the EfficientNet-B5 architecture showcases the potential for enhanced generalization and adaptability. Our results

establish a new benchmark in cross-dataset evaluation, underscoring the efficacy of our approach in combating deepfake threats across diverse datasets.

Despite these advancements, several directions for future research remain. First, enhancing the robustness of FSBI against adversarially generated deepfakes warrants exploration, as attackers continually evolve techniques to bypass detection methods. Second, investigating the integration of temporal features from video data could further improve detection accuracy by capturing inconsistencies over time. Additionally, extending the framework to handle emerging modalities, such as audio–visual deepfakes or 3D manipulations, could broaden its applicability.

CRediT authorship contribution statement

Ahmed Abul Hasanaath: Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Hamzah Luqman:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis. **Raed Katib:** Writing – original draft, Validation, Methodology. **Saeed Anwar:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

I have shared the link to my code in the attached article.

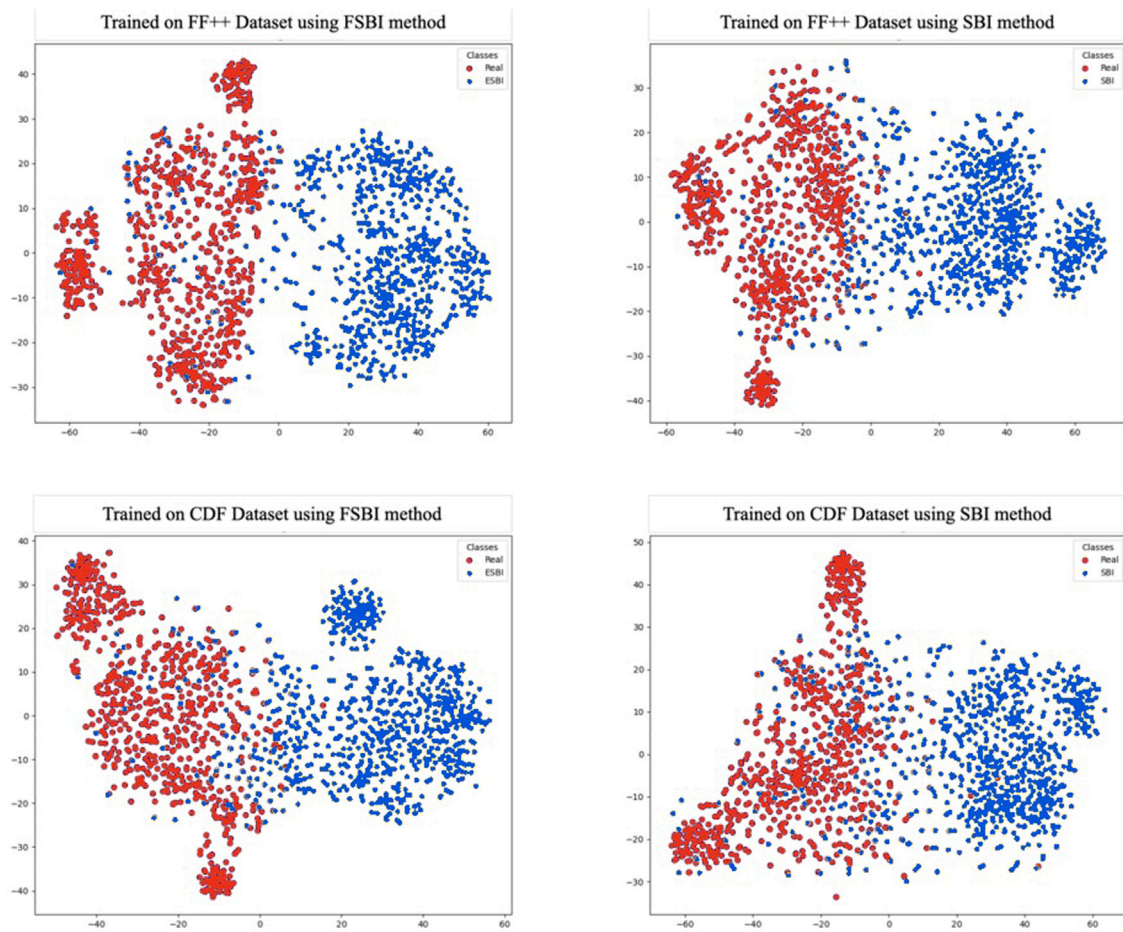


Fig. 4. Feature space visualization of FSBI (first column) and SBI (second column) trained on (a) FF++ and (b) Celeb-DF datasets.

References

- [1] M.S. Rana, M.N. Nobi, B. Murali, A.H. Sung, Deepfake detection: A systematic literature review, *IEEE Access* 10 (2022) 25494–25513, Conference Name: IEEE Access.
- [2] D.P. Kingma, M. Welling, Auto-encoding variational bayes, 2013, arXiv preprint [arXiv:1312.6114](https://arxiv.org/abs/1312.6114).
- [3] D.P. Kingma, M. Welling, An introduction to variational autoencoders, *Found. Trends® Mach. Learn.* 12 (4) (2019) 307–392, <http://arxiv.org/abs/1906.02691> [cs, stat], [arXiv:1906.02691](https://arxiv.org/abs/1906.02691) [cs, stat].
- [4] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, Y. Bengio, Generative adversarial nets, in: *Advances in Neural Information Processing Systems*, vol. 27, Curran Associates, Inc., 2014.
- [5] R. Tolosana, R. Vera-Rodriguez, J. Fierrez, A. Morales, J. Ortega-Garcia, Deepfakes and beyond: A survey of face manipulation and fake detection, *Inf. Fusion* 64 (2020) 131–148.
- [6] T. Karras, T. Aila, S. Laine, J. Lehtinen, Progressive growing of gans for improved quality, stability, and variation, 2017, arXiv preprint [arXiv:1710.10196](https://arxiv.org/abs/1710.10196).
- [7] T. Karras, S. Laine, T. Aila, A style-based generator architecture for generative adversarial networks, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 4401–4410.
- [8] T. Miyato, T. Kataoka, M. Koyama, Y. Yoshida, Spectral normalization for generative adversarial networks, 2018, arXiv preprint [arXiv:1802.05957](https://arxiv.org/abs/1802.05957).
- [9] deepfakes, Faceswap, 2023-02-22, URL <https://github.com/deepfakes/faceswap>, original-date: 2017-12-19T09:44:13Z.
- [10] J. Thies, M. Zollhöfer, M. Nießner, Deferred neural rendering: Image synthesis using neural textures, *Acm Trans. Graph. (TOG)* 38 (4) (2019) 1–12.
- [11] J. Thies, M. Zollhöfer, M. Stamminger, C. Theobalt, M. Nießner, Face2Face: Real-time face capture and reenactment of RGB videos, 2020-07-29, [arXiv:2007.14808](https://arxiv.org/abs/2007.14808), [http://arxiv.org/abs/2007.14808](https://arxiv.org/abs/2007.14808) [cs], [arXiv:2007.14808](https://arxiv.org/abs/2007.14808) [cs].
- [12] Y. Li, S. Lyu, Exposing deepfake videos by detecting face warping artifacts, 2018, arXiv preprint [arXiv:1811.00656](https://arxiv.org/abs/1811.00656).
- [13] M. Sahla Habeeba, A. Lijiya, A.M. Chacko, Detection of deepfakes using visual artifacts and neural network classifier, in: *Innovations in Electrical and Electronic Engineering: Proceedings of ICEEE 2020*, Springer, 2021, pp. 411–422.
- [14] F. Matern, C. Riess, M. Stamminger, Exploiting visual artifacts to expose deepfakes and face manipulations, in: *2019 IEEE Winter Applications of Computer Vision Workshops, WACVW, IEEE*, 2019, pp. 83–92.
- [15] X. Yang, Y. Li, S. Lyu, Exposing deep fakes using inconsistent head poses, in: *ICASSP 2019-2019 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP, IEEE*, 2019, pp. 8261–8265.
- [16] D. Cozzolino, J. Thies, A. Rössler, C. Riess, M. Nießner, L. Verdoliva, Forensic-transfer: Weakly-supervised domain adaptation for forgery detection, 2018, arXiv preprint [arXiv:1812.02510](https://arxiv.org/abs/1812.02510).
- [17] X. Xuan, B. Peng, W. Wang, J. Dong, On the generalization of GAN image forensics, in: *Chinese Conference on Biometric Recognition*, Springer, 2019, pp. 134–141.
- [18] H. Zhao, W. Zhou, D. Chen, T. Wei, W. Zhang, N. Yu, Multi-attentional deepfake detection, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 2185–2194.
- [19] Z. Sun, Y. Han, Z. Hua, N. Ruan, W. Jia, Improving the efficiency and robustness of deepfakes detection through precise geometric features, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 3609–3618.
- [20] Y. Zhao, B. Liu, M. Ding, B. Liu, T. Zhu, X. Yu, Proactive deepfake defence via identity watermarking, in: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2023, pp. 4602–4611.
- [21] Y. Wang, C. Chen, N. Zhang, X. Hu, WATCHER: Wavelet-guided texture-content hierarchical relation learning for deepfake detection, *Int. J. Comput. Vis.* (2024) 1–22.
- [22] J. Zheng, Y. Zhou, X. Hu, Z. Tang, DT-TransUNet: A dual-task model for deepfake detection and segmentation, in: *Chinese Conference on Pattern Recognition and Computer Vision, PRCV, Springer*, 2023, pp. 244–255.
- [23] K. Shiohara, T. Yamasaki, Detecting deepfakes with self-blended images, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 18720–18729.
- [24] Y. Jeong, D. Kim, S. Min, S. Joe, Y. Gwon, J. Choi, BiHPF: Bilateral high-pass filters for robust deepfake detection, in: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2022, pp. 48–57.

- [25] B. Liu, B. Liu, M. Ding, T. Zhu, X. Yu, TI2Net: Temporal identity inconsistency network for deepfake detection, in: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2023, pp. 4691–4700.
- [26] L. Verdoliva, Media forensics and deepfakes: an overview, *IEEE J. Sel. Top. Sign. Proces.* 14 (5) (2020) 910–932.
- [27] T.T. Nguyen, Q.V.H. Nguyen, D.T. Nguyen, D.T. Nguyen, T. Huynh-The, S. Nahavandi, T.T. Nguyen, Q.-V. Pham, C.M. Nguyen, Deep learning for deepfakes creation and detection: A survey, *Comput. Vis. Image Underst.* 223 (2022) 103525.
- [28] İ. İlhan, E. Balı, M. Karaköse, An improved DeepFake detection approach with NASNetLarge CNN, in: *International Conference on Data Analytics for Business and Industry, ICDABI*, 2022, pp. 598–602.
- [29] A. Khormali, J.-S. Yuan, DFDT: An end-to-end DeepFake detection framework using vision transformer, *Appl. Sci.* 12 (6) (2022-01) 2953, Number: 6 Publisher: Multidisciplinary Digital Publishing Institute.
- [30] Y. Luo, Y. Zhang, J. Yan, W. Liu, Generalizing face forgery detection with high-frequency features, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 16317–16326.
- [31] Y. Xu, K. Raja, L. Verdoliva, M. Pedersen, Learning pairwise interaction for generalizable DeepFake detection, in: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2023, pp. 672–682.
- [32] T. Zhao, X. Xu, M. Xu, H. Ding, Y. Xiong, W. Xia, Learning self-consistency for deepfake detection, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 15023–15033.
- [33] A.V. Nadimpalli, A. Rattani, On improving cross-dataset generalization of deepfake detectors, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 91–99.
- [34] A. Jevnisek, S. Avidan, Aggregating layers for deepfake detection, in: *26th International Conference on Pattern Recognition, ICPR, IEEE*, 2022, pp. 2027–2033.
- [35] D.A. Coccomini, R. Caldelli, F. Falchi, C. Gennaro, G. Amato, Cross-forgery analysis of vision transformers and cnns for deepfake image detection, in: *Proceedings of the 1st International Workshop on Multimedia AI Against Disinformation*, 2022, pp. 52–58.
- [36] T. Wang, H. Cheng, K.P. Chow, L. Nie, Deep convolutional pooling transformer for deepfake detection, *ACM Trans. Multimed. Comput. Commun. Appl.* 19 (6) (2023) 1–20.
- [37] D.A. Coccomini, N. Messina, C. Gennaro, F. Falchi, Combining efficientnet and vision transformers for video deepfake detection, in: *International Conference on Image Analysis and Processing*, Springer, 2022, pp. 219–229.
- [38] D. Wodajo, S. Atnafu, Deepfake video detection using convolutional vision transformer, 2021, arXiv preprint [arXiv:2102.11126](https://arxiv.org/abs/2102.11126).
- [39] Y.-J. Heo, Y.-J. Choi, Y.-W. Lee, B.-G. Kim, Deepfake detection scheme based on vision transformer and distillation, 2021, arXiv preprint [arXiv:2104.01353](https://arxiv.org/abs/2104.01353).
- [40] S. Chen, T. Yao, Y. Chen, S. Ding, J. Li, R. Ji, Local relation learning for face forgery detection, 2021-05-06, [arXiv:2105.02577](https://arxiv.org/abs/2105.02577), [arXiv:2105.02577](https://arxiv.org/abs/2105.02577) [cs].
- [41] L. Chen, Y. Zhang, Y. Song, L. Liu, J. Wang, Self-supervised learning of adversarial example: Towards good generalizations for deepfake detection, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 18710–18719.
- [42] S. Lee, S. Tariq, J. Kim, S.S. Woo, Tar: Generalized forensic framework to detect deepfakes using weakly supervised learning, in: *IFIP International Conference on ICT Systems Security and Privacy Protection*, Springer, 2021, pp. 351–366.
- [43] W. Ge, J. Patino, M. Todisco, N. Evans, Explaining deep learning models for spoofing and deepfake detection with SHapley additive exPlanations, 2021-10-07, [arXiv:2110.03309](https://arxiv.org/abs/2110.03309), [arXiv:2110.03309](https://arxiv.org/abs/2110.03309) [eess].
- [44] Q. Yin, W. Lu, B. Li, J. Huang, Dynamic difference learning with spatio-temporal correlation for deepfake video detection, *IEEE Trans. Inf. Forensics Secur.* (2023).
- [45] H. Wang, Z. Liu, S. Wang, Exploiting complementary dynamic incoherence for deepfake video detection, *IEEE Trans. Circuits Syst. Video Technol.* 33 (8) (2023) 4027–4040.
- [46] F. Becattini, C. Bisogni, V. Loia, C. Pero, F. Hao, Head pose estimation patterns as deepfake detectors, *ACM Trans. Multimed. Comput. Commun. Appl.* 20 (11) (2024) 1–24.
- [47] Y. Ju, S. Hu, S. Jia, G.H. Chen, S. Lyu, Improving fairness in deepfake detection, in: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2024, pp. 4655–4665.
- [48] L. Lin, X. He, Y. Ju, X. Wang, F. Ding, S. Hu, Preserving fairness generalization in deepfake detection, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 16815–16825.
- [49] K. Xu, X. Hu, X. Zhou, X. Xu, L. Qi, C. Chen, RLGC: Reconstruction learning fusing gradient and content features for efficient deepfake detection, *IEEE Trans. Consum. Electron.* (2024).
- [50] L. Li, J. Bao, T. Zhang, H. Yang, D. Chen, F. Wen, B. Guo, Face X-ray for more general face forgery detection, 2019, [arXiv:1912.13458](https://arxiv.org/abs/1912.13458).
- [51] M. Tan, Q. Le, Efficientnet: Rethinking model scaling for convolutional neural networks, in: *International Conference on Machine Learning*, PMLR, 2019, pp. 6105–6114.
- [52] A. Rossler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, M. Nießner, Faceforensics++: Learning to detect manipulated facial images, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 1–11.
- [53] Y. Li, X. Yang, P. Sun, H. Qi, S. Lyu, Celeb-df: A large-scale challenging dataset for deepfake forensics, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 3207–3216.
- [54] GitHub - deepfakes/faceswap: Deepfakes software for all — [github.com](https://github.com/deepfakes/faceswap), 2024, <https://github.com/deepfakes/faceswap>. (Accessed 07 March 2024).
- [55] J. Thies, M. Zollhofer, M. Stamminger, C. Theobalt, M. Nießner, Face2face: Real-time face capture and reenactment of RGB videos, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 2387–2395.
- [56] J. Thies, M. Zollhöfer, M. Nießner, Deferred neural rendering: Image synthesis using neural textures, *Acmm Trans. Graph. (TOG)* 38 (4) (2019) 1–12.
- [57] GitHub - MarekKowalski/FaceSwap: 3D face swapping implemented in Python — [github.com](https://github.com/MarekKowalski/FaceSwap), 2023, <https://github.com/MarekKowalski/FaceSwap/>. (Accessed 12 November 2023).
- [58] A. Jevnisek, S. Avidan, Aggregating layers for deepfake detection, in: *2022 26th International Conference on Pattern Recognition, ICPR, IEEE*, 2022, pp. 2027–2033.
- [59] D.A. Coccomini, N. Messina, C. Gennaro, F. Falchi, Combining efficientnet and vision transformers for video deepfake detection, in: *International Conference on Image Analysis and Processing*, Springer, 2022, pp. 219–229.
- [60] Y. Xu, K. Raja, L. Verdoliva, M. Pedersen, Learning pairwise interaction for generalizable DeepFake detection, in: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2023, pp. 672–682.
- [61] D. Wodajo, S. Atnafu, Deepfake video detection using convolutional vision transformer, 2021, arXiv preprint [arXiv:2102.11126](https://arxiv.org/abs/2102.11126).
- [62] Y. Luo, Y. Zhang, J. Yan, W. Liu, Generalizing face forgery detection with high-frequency features, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 16317–16326.
- [63] T. Zhao, X. Xu, M. Xu, H. Ding, Y. Xiong, W. Xia, Learning self-consistency for deepfake detection, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 15023–15033.
- [64] A. Haliassos, K. Vougioukas, S. Petridis, M. Pantic, Lips don't lie: A generalisable and robust approach to face forgery detection, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 5039–5049.
- [65] Y. Zheng, J. Bao, D. Chen, M. Zeng, F. Wen, Exploring temporal coherence for more general video face forgery detection, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 15044–15054.
- [66] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 770–778.
- [67] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, L.-C. Chen, Mobilenetv2: Inverted residuals and linear bottlenecks, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 4510–4520.